

Vikhram S

Prospective MS/PhD Researcher | Vision–Language Models | Explainable AI | Multimodal Intelligence

[in linkedin.com/in/vikhram-s](https://www.linkedin.com/in/vikhram-s) github.com/Vikhram-S [@ vikhramselvacumaran@gmail.com](mailto:vikhramselvacumaran@gmail.com) [vikhrm-s.github.io](https://vikhram-s.github.io)
[Google Scholar](#)

Researcher in artificial intelligence focusing on multimodal learning, explainability, and retrieval-augmented reasoning. First-author publication at *International Conference on Innovative Computing and Communication (ICICC 2026, Springer Lecture Notes in Networks and Systems)* and contributor to a UN Women–recognized national AI initiative. Develops vision–language and reasoning systems, with open-source tools used at scale (35,000+ downloads). Interested in interpretable AI for high-stakes settings.

RESEARCH INTERESTS

Core Interests	Multimodal AI, Vision–Language Models, Explainable AI (XAI), Scientific AI Systems, Constitutional AI, AI Governance
Applied Interests	Public Interest Technology, Retrieval-Augmented Generation, Healthcare Intelligence, Technology Policy & Digital Governance

EDUCATION

2026	Saveetha Engineering College	Chennai, India
2022	B.E. in Electronics and Communication Engineering CGPA: 8.10/10.0. Coursework: Machine Learning, Deep Learning, Digital Signal Processing, Computer Vision, Natural Language Processing, Probability & Statistics, Data Structures and Algorithms. Culminated in a peer-reviewed first-author publication (Springer LNNS, ICICC 2026), a B.E. thesis graded O and awarded Best Project, Dept. of ECE , and an internationally recognized applied-AI project developed with UN Women and India's MeitY.	

RESEARCH & INDUSTRY EXPERIENCE

Present	Independent Researcher – NariRaksha-3B & IndicConBench	Chennai, India
2026	Extended the published NariRaksha case study (India AI Impact Summit 2026 Compendium, MeitY & UN Women) into NariRaksha-3B , a QLoRA fine-tune of Qwen2.5-3B-Instruct on a purpose-built 100K-row dataset of Indian women's-safety scenarios spanning risk classification, structured reasoning, and applicable legal context. - Independently designed the dataset, training methodology, and evaluation protocol; released with full training methodology and a documented evaluation roadmap as a reproducible baseline for follow-on research. - Actively maintains and extends IndicConBench, a bilingual (English/Hindi) benchmark of 8,400+ examples spanning QA, retrieval, reasoning, and classification tasks, released as a public dataset on Hugging Face.	
May 2026	Virtusa Consulting Services Pvt Ltd 🌐 Jatayu Hackathon 2026, Agentic AI Track	Chennai, India
March 2026	Intern - Architected and shipped CodePerfect Auditor, a healthcare revenue-integrity platform that automatically reconciles clinical documentation against billed ICD-10/CPT codes, surfacing every discrepancy through a traceable, evidence-linked audit graph. - Engineered a clinical NLP extraction pipeline and a configurable rules engine for automated code-mapping and discrepancy detection with quantified revenue-impact scoring. - Delivered a production-grade full-stack system (FastAPI backend, Streamlit interface) from concept to working demo within a competitive hackathon timeline, selected among the top agentic-AI submissions.	
2026	Saveetha Engineering College 🌐 Dept. of Electronics and Communication Engineering	Chennai, India
2025	Lead Researcher, ExplainableVLM-Rad (B.E. Thesis) Advisor: Dr. J. Jeffin Gracewell - Designed and built a ViT + BioGPT vision-language pipeline generating clinically coherent chest X-ray reports with natively integrated visual and textual explainability, trained and evaluated on MIMIC-CXR (227,835 images), IU X-Ray, and CheXpert. - Outperformed CNN–RNN and Transformer baselines across all linguistic and clinical metrics – BLEU-4: 0.489, ROUGE-L: 0.521, CheXbert F1: 0.521 – reducing hallucinated findings by 28% relative to baseline on a 1,000-image validation set. - Achieved a Grad-CAM IoU of 0.42, token–patch alignment of 0.71, and an 87% radiologist explainability satisfaction rating, cutting simulated report-interpretation time by 31%. - Graded O (viva voce examined); awarded Best Project, Dept. of ECE; manuscript adapted for submission to a Q1 Scopus/Web of Science-indexed journal.	

2026 2025	Saveetha Engineering College 🌐 Dept. of Electronics and Communication Engineering Chennai, India Research Assistant & Team Lead, Corresponding Author Advisor: Dr. J. Jeffin Gracewell ➤ Led a 3-person research team as corresponding author on a first-author paper ranked in the top 15% of submissions at ICICC 2026 (Springer LNNS) – an explainable SLM-guided vision–language model for multi-class skin lesion recognition. ➤ Directed the team through the complete applied-AI research lifecycle: problem formulation, model architecture, experimentation, and manuscript preparation. ➤ Investigated cross-modal architectures integrating perception, retrieval, and language reasoning for high-stakes, domain-specific decision-making systems.	
Present 2025	Saveetha Engineering College 🌐 Tech Society Invited Advisor Appointed by <i>Mr. C. Obed Otto, Dean – ICT</i> ➤ Provides technical leadership within Tech Society, a peer-led community for collaborative learning, innovation, and skill development across technology domains. ➤ Architected and delivered 4+ advanced technical curricula spanning AI Agents, Quantum Machine Learning, Financial ML, and open-source AI tooling, upskilling 100+ student researchers on deployment-oriented ML workflows.	Chennai, India

RESEARCH PUBLICATIONS

- [1] **Explainable SLM-Guided Vision–Language Model for Multi-class Skin Lesion Recognition.** J. J. Gracewell, J. Yuvaraj, and S. Vikhram. *Proceedings of the International Conference on Innovative Computing and Communication (ICICC 2026)*, Lecture Notes in Networks and Systems, **Springer Nature Switzerland**, pp. 611–622. Scopus, DBLP, zbMATH, EI Compindex indexed; ranked top 15% of reviewed submissions. doi.org/10.1007/978-3-032-30310-3_49
- [2] **NariRaksha: Gender-Responsive AI for Women’s Safety.** S. Vikhram and J. J. Gracewell. *India AI Impact Summit 2026 Compendium – Real-World Impact of AI and Gender Empowerment: Casebook on AI and Gender Empowerment*, pp. 68–70. Ministry of Electronics and Information Technology (MeitY), Ministry of Women and Child Development (MWCD), and IndiaAI Mission, released in partnership with **UN Women**. **1 of 23 case studies selected globally from 235+ submissions across 50+ countries.**
- [3] **Explainable Vision-Language Generative Models for Radiology Report Image Synthesis and Interpretation.** S. Vikhram, J. J. Gracewell, and J. Yuvaraj. *Manuscript in preparation*, adapted from B.E. thesis work graded O and awarded Best Project, targeting a Q1 Scopus/Web of Science-indexed journal.

HONORS AND AWARDS

1. Published Contributor, India AI Impact Summit 2026 (MeitY and UN Women) – lead author of a case study selected among 23 from 235+ global submissions across 50+ countries; individually recognized by senior UN Women leadership at the official publication launch February 2026
2. Springer LNNS Publication Recognition, ICICC 2026 – first-author paper flagged in the official acceptance notification as among the top 15% of all papers reviewed; indexed by Scopus, DBLP, zbMATH, EI Compindex, and SCImago February 2026
3. Best Project Award, Dept. of ECE, Saveetha Engineering College – awarded for B.E. thesis “Explainable Vision-Language Generative Models for Radiology Report Image Synthesis and Interpretation” (graded O; Rs. 3,000 prize) 2026
4. Invited Advisor, Machine Learning Tech Society – appointed by the Dean, ICT, Saveetha Engineering College, for academic and technical leadership; designed and delivered programs on AI Agents, Financial ML, Quantum ML, and Open Source AI 2025
5. Paryavaran Ratna Award – conferred by the Indian Centre for Wildlife and Environmental Studies, South Asia Region, for contributions to environmental stewardship and sustainability initiatives 2018
6. Unique Performance Award for Academic Excellence – Shri Vidhyaa Giri Matriculation Higher Secondary School 2017
7. Distinction, Universal Talent Search Examination – Shakespeare Institute of English Studies 2016

INVITED TALKS AND LECTURES

1. Cyber Law and Compliance Series: Security, Contracts, and Intellectual Property in the Digital Era – Dept. of Computer Science and Engineering (Cyber Security), Saveetha Engineering College October 24, 2025
2. Software Engineering Practices for Reproducible Deep Learning Systems – Dept. of Computer Science and Engineering, Saveetha Engineering College September 11, 2025

3. Financial Machine Learning: Macroeconomic Indicators, Forecasting, and Predictive Analytics – Dept. of Artificial Intelligence and Machine Learning, Saveetha Engineering College
May 16, 2025
4. Quantum Machine Learning: Emerging Directions in Hybrid Quantum–Classical AI – Dept. of Artificial Intelligence and Machine Learning, Saveetha Engineering College
May 16, 2025
5. AI Agents and Open Source Ecosystems for Intelligent Software Development – Dept. of Artificial Intelligence and Data Science, Saveetha Engineering College
March 1, 2025

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C, Java
ML/DL Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, Unsloth
AI Specializations	Vision–Language Models (VLMs), Vision Transformers (ViT), Large/Small Language Models (LLMs/SLMs), Retrieval-Augmented Generation (RAG), Explainable AI (XAI), QLoRA Fine-Tuning, Prompt Engineering
Domains	Multimodal Learning, Medical Imaging AI, Clinical NLP, Legal NLP, Computer Vision, Model Evaluation & Interpretability
Backend / MLOps	FastAPI, Streamlit, REST API Design, Git/GitHub, Docker, Model Deployment, Hugging Face Hub, PyPI Publishing

GLOBAL ENGAGEMENT AND CERTIFICATIONS

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1. Individually recognized by senior UN Women leadership at the official India AI Impact Summit 2026 compendium launch for contributions to gender-responsive, public-interest AI.
 2. Engaged directly with senior AI industry, research, and policy leadership, including Calista Redmond (VP, Global AI Initiatives, NVIDIA), Pradeep Gupta (VP, Solutions Architecture, NVIDIA), John Fanelli (VP, Enterprise Software, NVIDIA), Dr. Jacki O’Neill (Director, Microsoft Research Africa), Dr. Aisha Walcott-Bryant (Head of Research, Google Africa), Christine Arab (Regional Director, UN Women Asia Pacific), Prof. Vukosi Marivate (Professor of Computer Science, University of Pretoria), and Adarsh Kumar (World Bank).
 3. Created and maintains IndianConstitution, an open-source legal NLP package for semantic search over India’s constitutional text, surpassing 33,000+ cumulative PyPI downloads, and its bilingual extension IndicConBench (8,400+ QA/retrieval/reasoning examples, released on Hugging Face).
 4. Deep Learning Specialization, DeepLearning.AI (Andrew Ng) – Neural Networks, CNNs, RNNs, Sequence Models, Hyperparameter Tuning.
 5. Google Cybersecurity Specialization, Google – Credential ID: 3VZ635P8BB58.
 6. Dynamic Public Speaking, University of Washington (Coursera) – Credential ID: UHK2J067KIKI.

REFERENCES

Dr. J. Jeffin Gracewell <i>Associate Professor, DEPT. OF ECE</i> @ jeffingracewellj@saveetha.ac.in	Dr. Srigitha S. Nath <i>Professor & Head, DEPT. OF ECE</i> @ srigithasnath@saveetha.ac.in
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